Unconfounded but inflated causal estimates

Mathematical derivation

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I Setting

I. 1 Objectives

The objective of my article is to compare "biases" resulting from exaggeration in causal identification strategies on the one hand and from confounders in "naive" regression analysis on the other hand. This document aims to provide a mathematical derivation to illustrate this trade-off. Such a derivation may provide some intuition regarding the various mechanisms at play. It does not aim in any way to provide a general proof but merely a specific one, tied to the setting considered. To do so, I build on a simple and commonly used framework. While I keep the setting as general as possible, I have to make certain assumptions, in particular regarding the data generating process considered.

The objective in this mathematical derivation is to compare the exaggeration ratio for a naive regression with omitted variable bias (OVB) to that of the regression in which we control for the omitted variable (CTRL), of an Instrumental Variable (IV) model in which we instrument the variable affected by the omitted variable and of a reduced form (RED) estimation in which we directly regress the outcome on the instrument. Roughly, we aim to derive the conditions under which the exaggeration due to the loss in precision is greater than the omitted variable bias. The objective is to compare:

$$\mathbb{E}\left[\hat{\beta}_{\text{OVB}} |\ \beta_1, \sigma_{\text{OVB}}, \hat{\beta}_{\text{OVB}} > z_\alpha \sigma_{\text{OVB}}\right] \qquad \text{and} \qquad \mathbb{E}\left[\hat{\beta}_{..} |\ \beta_1, \sigma_{..}, \hat{\beta}_{..} > z_\alpha \sigma_{..}\right]$$

where .. stands for the controlled regression (CTRL), the IV (IV) and the reduced form (RED). β_1 represents the true effect of interest, $\hat{\beta}_{..}$ its estimate, $\sigma_{..}$ the standard error of the estimate and z_α the threshold value of a two-sided test of size α. In all of my analyses, I consider tests of size 5%. One can thus think of z_α as being equal to 1.96.

Note that we cannot simply compare the bias in the OVB case to the exaggeration resulting from the use of identification strategies and have to compare exaggeration ratios to exaggeration ratios. As we will see, this comparison is not straightforward as the bias affects the exaggeration in non linear ways [1].

In this document, I first describe the data generating process. I then derive a general equation for the exaggeration of a normally distributed estimator and discus how it evolves with the mean and standard deviation of the estimator. In a third section, I build an hand-wavy illustration of the confounding / exaggeration trade-off when we control for the omitted variable. Since most causal identification strategies can be interpreted as "controlling" for confounders (*e.g.* the control function (CF) approach for the IV), the CTRL case neatly summarizes the intuition for all strategies. In the last section, I build more formal proofs on the existence of the trade-off for the control approach, the IV and event-study estimated via 2SLS and the event-study estimated as a reduced form. For each strategy, I analyze how exaggeration evolves as a function of the parameters of interests. I also discuss how for each strategy, the main parameter of interest is linked the proportion of the variance in the explanatory variable of interest retained for identification.

1. The bias affects the average of statistically significant estimates in a linear way by shifting the distribution but also enters the Inverse Mills Ratio since statistically significant are estimates that are 2 s.d. + **bias** away from zero.

I. 2 Data generating process

Consider the usual linear regression model to with an omitted variable. We limit the number of covariates considered to a constant, an observed variable and an omitted variable. For all individual $i \in \{1, ..., n\}$, assume that:

$$y_i = \beta_0 + \beta_1 x_i + \delta w_i + u_i \tag{1}$$

where $x_i \in \mathbb{R}$ is the observed explanatory variable of interest, $y_i \in \mathbb{R}$ the observed outcome of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$ an unobserved error term. $(\beta_0, \beta_1, \delta) \in \mathbb{R}^3$ are unknown parameters. β_1 is the parameter of interest. Treatment effects are homogeneous ($\forall i \in \{1, ..., n\},\ \beta_{1i} = \beta_1 = \text{cst}$). This equation can be rewritten as:

$$y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i$$

with $\boldsymbol{\beta} \in (\beta_0, \beta_1)'$ and $\mathrm{x}_i = (1, x_i)'$. For readability, we introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$.

We further assume that the usual OLS assumptions hold:
— $(y_i, \mathrm{x}_i, w_i, u_i)$ are iid,
— Finite second moments, *ie* $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E}[w_i^2] < \infty$, $\mathbb{E}[x_i^2] < \infty$,
— Positive-definiteness of $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']$,
— u_i is conditional mean-zero, *ie* $\mathbb{E}[u_i | x_i, w_i] = 0$, and is uncorrelated with x_i and w_i, *ie* $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[w_i u_i] = 0$.

Note that conditional mean-zero implies that $\mathbb{E}[u_i | x_i] = 0$ (since $\mathbb{E}[u_i | x_i] = \mathbb{E}[\mathbb{E}[u_i | x_i, w_i | x_i]] = \mathbb{E}[0 | x_i] = 0$). We further assume homoskedasticity such that $\mathbb{E}[u_i^2 | x_i, w_i] = \sigma_u^2$ is constant. It implies that $\mathbb{E}[u_i^2 | x_i] = \sigma_u^2$ (since $\mathbb{E}[u_i^2 | x_i] = \mathbb{E}[\mathbb{E}[u_i^2 | x_i, w_i] | x_i] = \mathbb{E}[\sigma_u^2 | x_i] = \sigma_u^2$).

We however assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, we further assume that the unobserved variable is centred, *ie* $\mathbb{E}[w_i] = 0$. We also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^{\perp}$) does not vary with x_i. Thus, $\mathrm{Var}(w_i^{\perp} | x_i) = \mathrm{Var}(w_i^{\perp})$. We assume the following data generating process for x_i:

$$x_i = \mu_x + \gamma w_i + \epsilon_i \tag{2}$$

I. 3 Summary of the key assumptions

To ease the use of assumptions in the following sections, we summarize some that we will explicitly mobilize:
— $y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i$
— $x_i = \mu_x \gamma w_i + \epsilon_i$
— $\mathbb{E}[u_i | x_i, w_i] = 0$ and $\mathbb{E}[u_i | x_i] = 0$
— $\mathbb{E}[u_i^2 | x_i, w_i] = \sigma_u^2$ and $\mathbb{E}[u_i^2 | x_i] = \sigma_u^2$
— $\mathbb{E}[w_i] = 0$

— $\text{Var}(w_i^{\perp}|x_i) = \text{Var}(w_i^{\perp})$
— $\beta_1 > 0$ and $\delta\gamma > 0$ (the latter implies that the OVB is an upward bias).

II Exaggeration ratio and its properties

I first define the exaggeration ratio, then compute it for a normally distributed estimator and study its general properties. This then enables me to illustrate the intuition behind the link between imprecision and exaggeration.

Definition 1. *For an estimator $\hat{\beta}$ of a true effect of magnitude β_1, I define the exaggeration ratio E as the ratio expected value of the estimate conditional on the estimate being statistically significant over the true effect:*

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\mathbb{E}\left[\hat{\beta}|\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]}{\beta_1} \tag{3}$$

Note that Gelman & Carlin (2014) consider the expected value of the *absolute value* of the estimate: $\mathbb{E}\left[|\hat{\beta}||\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]/\beta_1$. To simplify the derivation, I abstract from the absolute value. My exaggeration ratio therefore depends on the well-known expected value of a truncated normal distribution instead of on the expected value of a truncated *folded* normal distribution.

Also note that, in this paper, we focus on estimators that can be decomposed into $\beta = \beta_1 + b$, with β_1 the true effect and b a bias independent of β_1. Since the objective of this project is mostly to analyze when the exaggeration is greater than the OVB, it makes sense to focus on cases in which they bias the estimate in the same direction *i.e.* β_1 and b of the same sign. I focus on the case where the true effect is positive and the OVB upward, *i.e.*, $\beta_1 > 0$ and $b \geq 0$. The reasoning for negative true effect and bias are is analogous but with signs flipped.

Finally, note that in this document, I refer to E as the "exaggeration ratio" and $\mathbb{E}\left[\hat{\beta}|\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right]$ as "exaggeration" to avoid using the two terms interchangeably.

II. 1 A general expression for the exaggeration ratio

Lemma 1. *For an estimator $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$ of a true effect of magnitude $\beta_1 > 0$ and such that $\beta = \beta_1 + b$,*

$$E(\beta, \sigma, \beta_1, z_\alpha) = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) \tag{4}$$

Proof. This exaggeration ratio is can be derived directly using properties of a truncated normal or directly using the expression of the inverse Mills ratio. Yet, since this document is a thorough description of the analysis, I derive this expression analytically.

Since $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2)$, its support is $\mathbb{R}$ and pdf is $f_{\hat{\beta}} : x \mapsto \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\beta)^2}{2\sigma^2}}$. Thus, by definition of the

conditional expectation,

$$\mathbb{E}\left[\hat{\beta}|\ \beta_1, \sigma, \hat{\beta} > z_\alpha \sigma\right] = \frac{1}{\Pr(\hat{\beta} > z_\alpha \sigma)} \int_{z_\alpha \sigma}^{\infty} x \times f_{\hat{\beta}}(x) \mathrm{d}x$$

Now,

$$\begin{aligned}
\Pr(\hat{\beta} > z_\alpha \sigma) &= \int_{z_\alpha \sigma}^{\infty} f_{\hat{\beta}}(x) \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha \sigma}^{\infty} e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t && \text{by the change of variable } x = \sigma t + \beta \\
&= 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) && \text{with } \Phi \text{ the cdf of the std. normal} \\
&= \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) && \text{since } \forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)
\end{aligned}$$

and,

$$\begin{aligned}
\int_{z_\alpha \sigma}^{\infty} x \times f_{\hat{\beta}}(x) \mathrm{d}x &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{z_\alpha \sigma}^{\infty} x e^{-\frac{(x-\beta)^2}{2\sigma^2}} \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} (\sigma t + \beta) e^{-\frac{t^2}{2}} \mathrm{d}t && \text{by the same change of variable} \\
&= \beta \times \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} e^{-\frac{t^2}{2}} \mathrm{d}t + \sigma \frac{1}{\sqrt{2\pi}} \int_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} t e^{-\frac{t^2}{2}} \mathrm{d}t \\
&= \beta \left(1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right)\right) + \sigma \frac{1}{\sqrt{2\pi}} \left[-e^{-\frac{t^2}{2}}\right]_{z_\alpha - \frac{\beta}{\sigma}}^{\infty} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \frac{1}{\sqrt{2\pi}} e^{-\frac{(z_\alpha - \beta)^2}{2\sigma^2}} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \phi\left(z_\alpha - \frac{\beta}{\sigma}\right) && \text{with } \phi \text{ the pdf of the std. norm.} \\
&= \beta \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right) + \sigma \phi\left(\frac{\beta}{\sigma} - z_\alpha\right) && \text{since } \forall x \in \mathbb{R},\ \phi(-x) = \phi(x)
\end{aligned}$$

Thus, we have,

$$E(\beta, \sigma, \beta_1, z_\alpha) = \frac{\beta}{\beta_1} + \frac{\sigma}{\beta_1} \frac{\phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}{\Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)}$$

where ϕ and Φ are respectively the pdf and the cdf of the standard normal distribution. We then introduce the inverse Mills ratio, the function $\lambda : x \mapsto \phi(x)/\Phi(x)$. In addition, since we consider an estimator whose mean can be decomposed in $\beta = \beta_1 + b$, with β_1 the true effect and b a bias

independent of β_1, we can then write the exaggeration ratio as:

$$\boxed{E(\beta, \sigma, \beta_1, z_\alpha) = 1 + \frac{b}{\beta_1} + \frac{\sigma}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)}$$

□

Another important metric in this paper is statistical power. It defined as the probability of rejecting the null of no effect when it is actually true. Hence, for a usual two-sided t-test with $n \gg k$, with k the number of covariates, the statistical power p is given by:

$$\begin{aligned} p(\beta, \sigma, \beta_1, z_\alpha) &= \Pr(\hat{\beta} < -z_\alpha\sigma) + \Pr(\hat{\beta} > z_\alpha\sigma) \\ &= \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} < -z_\alpha - \frac{\beta}{\sigma}\right) + \Pr\left(\frac{\hat{\beta} - \beta}{\sigma} > z_\alpha - \frac{\beta}{\sigma}\right) \\ &= \Phi\left(-z_\alpha - \frac{\beta}{\sigma}\right) + 1 - \Phi\left(z_\alpha - \frac{\beta}{\sigma}\right) \qquad \text{since } \frac{\hat{\beta} - \beta}{\sigma} \sim \mathcal{N}(0, 1) \end{aligned}$$

And thus, since $\forall x \in \mathbb{R},\ \Phi(-x) = 1 - \Phi(x)$,

$$\boxed{p(\beta, \sigma, \beta_1, z_\alpha) = 1 - \Phi\left(\frac{\beta}{\sigma} + z_\alpha\right) + \Phi\left(\frac{\beta}{\sigma} - z_\alpha\right)} \tag{5}$$

II. 2 Evolution of the exaggeration ratio with σ, β and b

Corollary 1. *For $\beta_1 > 0$ and $b > 0$, the exaggeration ratio increases with σ, the standard error of the estimator and decreases with β_1, the true effect size. It thus decreases with the Signal to Noise Ratio (SNR) of the estimator $\frac{\beta_1}{\sigma}$.*

Proof. From equation 4, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$,

$$\frac{\partial E}{\partial \sigma}(\beta, \sigma, \beta_1, z_\alpha) = \frac{1}{\beta_1}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) - \frac{\beta_1 + b}{\sigma}\lambda'\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)$$

Since the inverse Mills ratio being a positive, decreasing function, $\forall x \in \mathbb{R}, \lambda(x) > 0$ and $\lambda'(x) < 0$. Thus, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$, $\frac{1}{\beta_1}\lambda\left(\frac{\beta_1+b}{\sigma} - z_\alpha\right) > 0$ and $-\frac{\beta_1+b}{\sigma}\lambda'\left(\frac{\beta_1+b}{\sigma} - z_\alpha\right) > 0$ and $\dfrac{\partial E}{\partial \sigma} > 0$. <u>The exaggeration ratio is an increasing function of σ, the variance of the estimator</u>.

Similarly, from equation 4, $\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$,

$$\frac{\partial E}{\partial \beta_1}(\beta, \sigma, \beta_1, z_\alpha) = -\frac{b}{\beta_1^2} + \frac{1}{\beta_1}\lambda'\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right) - \frac{\sigma}{\beta_1^2}\lambda\left(\frac{\beta_1 + b}{\sigma} - z_\alpha\right)$$

$\forall(\beta, \sigma, \beta_1, z_\alpha) \in \mathbb{R}_+^4$, each term of this partial derivative is negative and thus $\dfrac{\partial E}{\partial \sigma} < 0$. <u>The exaggeration ratio is a decreasing function of β_1, the true effect size</u>.

Combining these two results implies that the exaggeration ratio is a decreasing function of β_1/σ, the signal to noise ratio. □

Note that these two results have been proven by Lu et al. (2019) and Zwet & Cator (2021) in the case of an exaggeration defined as in Gelman & Carlin (2014), *i.e.*, as the expected value of the absolute value of the point estimate.

Figure 1 gives an intuition for the relationship between the precision of an estimator and exaggeration by ploting the distribution of two unbiased estimators (that therefore have identical means) but different variances and distinguishing between estimates that are and are not significant.

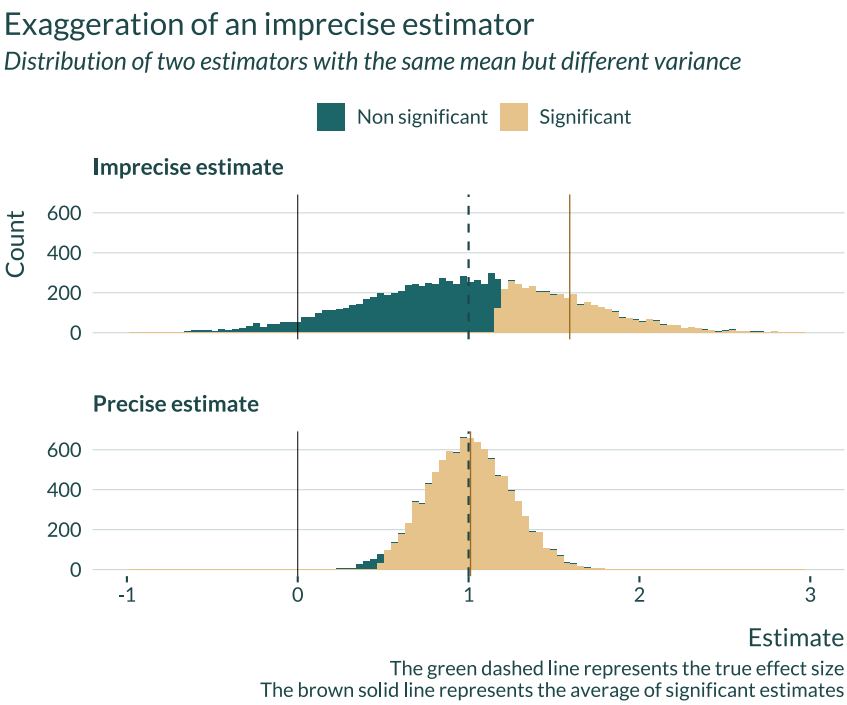

Figure 1 – Intuition for exaggeration of imprecise estimates

In the case of the precise estimator, the mean of statistically significant estimates is almost equal to the true effect. The estimator is unbiased and almost all estimates are significant. However, in the case of the imprecise estimator, only a sub-sample of the estimates are significant: those that are far away enough from 0 (at least 1.96 s.d.). As a consequence, the mean of statistically significant estimates is different from the true effect, even if the estimator is unbiased. This hints to the trade-off between biased estimates from naive regressions and less precise causal inference estimates.

III Intuition for controls and exaggeration

In this section, I give the intuition for why controlling for an omitted variable can actually cause exaggeration, depending on the characteristics of the variable one controls for. I then, based on this result, I discuss how using causal identification strategies can increase exaggeration.

III. 1 Controlling can increase exaggeration

III. 1.1 Controlling can increase variance...

In section II, we have seen that the exaggeration increases with the variance of a normally distributed estimator. Here, we will discuss under which conditions controlling for an omitted variable can increase the variance of the estimator and thus exaggeration. To do so, consider the DGP described in section I. 2 (recall that $y = \beta_0 + \beta_1 x + \delta w + u$ and $x = \mu_x + \gamma w + \epsilon$) and the two following regression models:

1. A "naive" regression of y on x (with w omitted):

$$y = \beta_{0,\text{OVB}} + \beta_{1,\text{OVB}} x + u_{\text{OVB}} \tag{6}$$

2. The controlled regression of y on x and w accurately represents the DGP:

$$y = \beta_{0,\text{CTRL}} + \beta_{1,\text{CTRL}} x + \delta_{\text{CTRL}} w + u_{\text{CTRL}} \tag{7}$$

Since we omitted w in equation 8, we can show the usual result such know that $\beta_{1,\text{OVB}} \neq \beta_1$ [2]. Similarly, we can easily show that $\beta_{1,\text{CTRL}} = \beta_1$. We know that the variance of $\beta_{1,\text{OVB}}$ is given by:

$$\text{Var}(\hat{\beta}_{1,\text{OVB}}) = \sigma^2_{\text{OVB}} = \frac{\text{Var}(u_{\text{OVB}})}{n\ \text{Var}(x)} = \boxed{\sigma^2_{\text{OVB}} = \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x)}}$$

where $y^{\perp x}$ is the component of y that is orthogonal to x. Using similar notation, and Frisch-Waugh-Lovell theorem, the variance of $\beta_{1,\text{CTRL}}$ is:

$$\text{Var}(\hat{\beta}_{1,\text{CTRL}}) = \sigma^2_{\text{CTRL}} = \frac{\text{Var}(u^{\text{CTRL}})}{n\ \text{Var}(x^{\perp w})} = \boxed{\sigma^2_{\text{CTRL}} = \frac{\text{Var}(y^{\perp x,w})}{n\ \text{Var}(x^{\perp w})}}$$

Based on these two equations, we can see that, under certain conditions, the variance of the estimator is greater when we control for the omitted variable than when we do not:

$$\begin{aligned}
\sigma^2_{\text{CTRL}} > \sigma^2_{\text{OVB}} \quad &\Leftrightarrow \quad \frac{\text{Var}(y^{\perp x,w})}{n\ \text{Var}(x^{\perp w})} > \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x)} \\
&\Leftrightarrow \quad \frac{\text{Var}(y^{\perp x,w})}{\text{Var}(y^{\perp x})} > \frac{\text{Var}(x^{\perp w})}{\text{Var}(x)} \\
&\Leftrightarrow \quad 1 - R^2_{y^{\perp x} \sim w} > 1 - R^2_{x \sim w} \\
&\Leftrightarrow \quad R^2_{y^{\perp x} \sim w} < R^2_{x \sim w}
\end{aligned}$$

where $R^2_{x\sim w}$ is the R^2 or coefficient of determination of the regression of x on w. This notation follows Cinelli & Hazlett (2020). $(1 - R^2_{x\sim w})$ and $\frac{\text{Var}(x^{\perp w})}{\text{Var}(x)}$ thus both represent the Fraction of the Variance in x Unexplained (FVU) by w.

2. We do so in equation 12, noting that in our case, the second componnent of $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient for x in the regression of w on x. Denoting it $\tilde{\gamma}$, we have $\beta_{1,\text{OVB}} = \beta_1 + \delta\tilde{\gamma} \neq \beta_1$. Note that, we have $\tilde{\gamma} = \frac{\text{cov}(w,x)}{\sigma^2_x} = \frac{\text{cov}(w,x)}{\sigma^2_w}\frac{\sigma^2_w}{\sigma^2_x} = \frac{\gamma\sigma^2_w}{\sigma^2_x}$ which yields the same result as in section IV. 1, *i.e.* $\beta_{1,\text{OVB}} = \beta_1 + \delta\gamma\frac{\sigma^2_w}{\sigma^2_x}$.

Hence, if the omitted variable explains a larger part of the variance of explanatory variable than of the variance of the explained variable, controlling for it will increase the variance of the estimator. Stated differently, controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), it will increase the variance of the estimator.

Note that since in general for an outcome variable y, $R^2 = \text{corr}\,(\hat{y}, y)^2$ and since both regressions considered in the two R^2 above are univariate, we can rewrite the previous equivalence as

$$\sigma^2_{\text{CTRL}} > \sigma^2_{\text{OVB}} \quad \Leftrightarrow \quad \text{corr}(w, y^{\perp x})^2 < \text{corr}(w, x)^2$$

If the omitted variable is more correlated with x than with the residual part of y ($y^{\perp x}$), controlling for it will increase the variance of the estimator of interest.

Unessentially, note that we could rewrite the variance of the two estimators using the R^2 notations:

$$\sigma^2_{\text{OVB}} = \frac{\sigma^2_y(1 - R^2_{y\sim x})}{n\ \sigma^2_x} \qquad \text{and} \qquad \sigma^2_{\text{CTRL}} = \frac{\sigma^2_y(1 - R^2_{y\sim x,w})}{n\ \sigma^2_x(1 - R^2_{x\sim w})}$$

III. 1.2 ... and exaggeration

Now that we have shown that the variance of the estimator can be greater when we control than when we do not, we will see that exaggeration can be greater when we control than when we do not. The following discussion is mostly illustrative. A formal proof is derived in the following sections as most results, while intuitive, are actually non trivial. This stems from the fact that while the exaggeration ratio increases with the variance, it is also affected in a non linear way by the bias. An estimator with a larger variance does not necessarily has a larger exaggeration if it also has a different bias.

Under the usual assumptions leading estimators to be normally distributed and noting that $b_{\text{CTRL}} = 0$, we can rewrite the exaggeration ratio in the case where we control for the omitted variable described in equation 4 as:

$$E_{\text{CTRL}} = 1 + \frac{\sigma_{\text{CTRL}}}{\beta_1}\lambda\left(\frac{\beta_1}{\sigma_{\text{CTRL}}} - z_\alpha\right)$$

If controlling for w absorbs most of the variation in x without absorbing much of the variation in $y^{\perp x}$ (*i.e.* $\text{Var}(x^{\perp w}) \ll \text{Var}(x)$ while $\text{Var}(y^{\perp x,w}) < \text{Var}(y^{\perp x})$), σ_{CTRL} can be arbitrarily large. Thus, we can compute the limit of E_{CTRL} when $\sigma_{\text{CTRL}} \to \infty$:

$$\lim_{\sigma_{\text{CTRL}}\to\infty} E_{\text{CTRL}} = \lim_{\sigma_{\text{CTRL}}\to\infty}\left(1 + \frac{\sigma_{\text{CTRL}}}{\beta_1}\lambda\left(-z_\alpha\right)\right) = +\infty$$

Controlling for a variable that absorbs most of the variation in x can cause exaggeration to be very large.

This result does not directly imply that controlling for this variable would lead E_{CTRL} to be larger than E_{OVB}. Indeed, E_{OVB} depends on the amount of omitted variable bias. In the following section, we however show that for any fixed amount of bias, exaggeration can be arbitrarily large, depending on

the parameter values. As a consequence, under some circumstances, exaggeration caused by the loss of precision can be greater that the omitted variable bias. This stems from the fact that the bias is $b_{\text{OVB}} = \delta\gamma\frac{\sigma_w^2}{\sigma_x^2}$. Thus, for any fixed amount of bias, γ can be large, if δ is relatively small. A large γ and a small δ mean that w is strongly correlated with x and weakly with y respectively. That implies a relatively large σ^2_{CTRL}, as discussed above and thus a large exaggeration. Thus, we can show that, in some cases, $E_{\text{CTRL}} > E_{\text{OVB}}$. Controlling "kills" the omitted variable bias but can decrease precision and thus lead publish estimates to be exaggerated.

III. 2 Causal identification strategies as control strategies

In this paper, I argue that causal identification strategies, although allowing to avoid confounders, can cause exaggeration (and that exaggeration can be greater than bias caused by confounders). The intuition behind this follows directly from the previous section. Causal identification strategies can be interpreted as controlling omitted variables. If this "control" absorbs too much variation in the treatment, based on the results from section III. 1, we can expect causal identification strategies to increase the risk of exaggeration.

Fixed-Effects (FEs) regression and FE-based identification strategies such as DiD constitute a good example of causal identification strategies as control strategies. We implement such strategies so that FEs control for the invariant, unobserved and arguably endogenous part of the variation. Now, following the reasoning built in section III. 1, if FEs absorb much more of the variation in the treatment than in the outcome, without annihilating a very large bias, exaggeration caused by controlling for fixed effects could be larger than the initial omitted variable bias.

The Control Function (CF) approach gives a straightforward interpretation of IV as a control strategy: we control for the predicted residuals of the regression of the endogenous variable of interest x on the instrument z. Based on section III. 1, if these residuals are more correlated with x than with $y^{\perp x}$, instrumenting will increase the variance of the estimator. Now, by construction and since these residuals are a part of x (the component of x that is not explained by z) they will not be correlated with $y^{\perp x}$ (the component of y that is orthogonal to x). They are however correlated with x and thus, the variance of the resulting CF estimator will be larger than the variance of the corresponding OLS. The less z explains of the variance in x, the larger the correlation between x and the predicted residuals and thus the larger the variance of the CF estimator and thus its exaggeration. In mathematical terms, and denoting $\hat{\epsilon}$ the predicted residuals of the regression of x on z, we have:

$$\sigma^2_{\text{OLS}} = \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x)} \quad \text{and} \quad \sigma^2_{\text{CF}} = \frac{\text{Var}(y^{\perp x,\hat{\epsilon}})}{n\ \text{Var}(x^{\perp\hat{\epsilon}})} = \frac{\text{Var}(y^{\perp x})}{n\ \text{Var}(x^{\perp\hat{\epsilon}})}$$

And thus,é

$$\frac{\sigma^2_{\text{OLS}}}{\sigma^2_{\text{CF}}} = \frac{\text{Var}(x^{\perp\hat{\epsilon}})}{\text{Var}(x)} = 1 - R^2_{x\sim\hat{\epsilon}} = R^2_{x\sim z}$$

The smaller the proportion of the variance of x that is explained by z ($R^2_{x\sim z}$), the larger the CF-variance, as compared to that of the OLS [3].

3. Note that all this reasoning is built under the assumption that no other control variables are included. We abstracted from them here since this section serves for illustrative purposes only.

I am not certain I see how I could build a similar argument for the RD, the event study an matching.

In this section, we have discussed the intuition behind the potential impacts of controlling, and thus using causal identification strategies, on exaggeration. Yet, since the bias enters the Inverse Mills Ratio, the comparison between the exaggeration ratios is not straightforward and cannot be done with simple arguments. We thus need to derive the actual distributions of the estimators to be able to accurately compute the exaggeration ratio.

IV Exaggeration ratios for each identification strategy

As discussed in section I. 1, the goal is to compare the exaggeration ratios in the case of the naive regression to that of the corresponding controlled regression, IV and reduced form. As discussed in section II, in order to compute the exaggeration ratios for the various estimators, we need to know their distributions. In this section, for each estimator we thus first derive its asymptotic distribution before computing its exaggeration ratio and studying how it evolves with the value of key parameters.

IV. 1 Naive regression (OVB)

In this section, we consider a "naive" regression of y on x (with w omitted). Since, we do not observe w, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + u_{\text{OVB}} \tag{8}$$

where $u_{\text{OVB}} = (u_1^{\text{OVB}}, ..., u_n^{\text{OVB}})' \in \mathbb{R}^n$ and by definition of the projection, $\mathbb{E}[X'u_{\text{OVB}}] = 0$. Note that we use capital letters to denote matrices (for instance $X = (\mathrm{x}_1', ..., \mathrm{x}_n')'$, with $\forall i, \mathrm{x}_i' = (1, x_i)$).

IV. 1.1 Asymptotic distribution

Lemma 2. *Based on the data generating process described in equation 1, for the naive OLS estimate $\hat{\beta}_{\text{OVB}}$ of the regression of y on x, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{OVB}} - \left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{\sigma_x^2}\right)$$

Proof. We first compute the bias of the estimator. From equation 8 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'u_{\text{OVB}} \\
\Rightarrow\quad & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'u_{\text{OVB}}]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'y] \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] \qquad \text{cf eq. 1} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta + \underbrace{\mathbb{E}[X'u]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned} \tag{9}$$

Since X and w are correlated and $\delta \neq 0$, $\boxed{\boldsymbol{\beta}_{\text{OVB}} \neq \boldsymbol{\beta}}$. $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient matrix of the projection of w on X. A similar derivation and discussion is carried out in Hansen (2022) for instance.

Let $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ denote the least square estimator of $\boldsymbol{\beta}_{\text{OVB}}$. We thus have:

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'y \tag{10}$$

$$\begin{aligned}
\Rightarrow\quad & \hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'(X\boldsymbol{\beta}_{\text{OVB}} + u_{\text{OVB}}) && \text{cf eq. 8} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X] = \boldsymbol{\beta}_{\text{OVB}} + (X'X)^{-1}\underbrace{\mathbb{E}[X'u_{\text{OVB}}]}_{0} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X]] = \boldsymbol{\beta}_{\text{OVB}} && \text{by the Law of Iterated Expect. (LIE)}
\end{aligned}$$

Thus, <u>$\hat{\boldsymbol{\beta}}_{\text{OVB}}$ is an unbiased estimator of $\boldsymbol{\beta}_{\text{OVB}}$ but a biased estimator of $\boldsymbol{\beta}$</u>. We then compute its asymptotic distribution.

We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i u_{\text{OVB},i}\right)$$

By the Weak Law of Large Numbers (WLLN), $\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Since $\mathbb{E}[\mathrm{x}_i u_{\text{OVB},i}] = 0$, by the Central Limit Theorem (CLT), $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i u_{\text{OVB},i}\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2])$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta} + \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}/n\right)} \tag{11}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x \\ x & x^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \text{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\underbrace{\text{var}(w_i)}_{\sigma_w^2} + \underbrace{\text{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{12}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]$, noting that:

$$\begin{aligned}
u_{\text{OVB},i} &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i'\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta\underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^{\perp} \qquad\qquad \text{where } w_i^{\perp} = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^{\perp})^2|x_i] \\
&= \mathbb{E}[u_i^2|x_i] + 2\delta\mathbb{E}[u_i w_i^{\perp}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + 2\delta\left(\mathbb{E}[u_i w_i|x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i|x_i]}_{0}\right) + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i\underbrace{\mathbb{E}[u_i|x_i, w_i]}_{0}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^{\perp})^2|x_i] = \text{Var}(w_i^{\perp}|x_i) + \mathbb{E}[w_i^{\perp}|x_i]^2$. Now, since $w_i^{\perp}$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^{\perp}|x_i] = 0$. And thus, since by assumption $\text{Var}(w_i^{\perp}|x_i) = \text{Var}(w_i^{\perp})$,

$$
\begin{aligned}
\mathbb{E}[(w_i^{\perp})^2|x_i] &= \mathrm{Var}(w_i^{\perp}|x_i) \\
&= \mathrm{Var}(w_i^{\perp}) \\
&= \mathbb{E}[(w_i^{\perp})^2] - \mathbb{E}[w_i^{\perp}]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x \underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 - 2\frac{\gamma^2\sigma_w^4}{\sigma_x^2} + \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2 - \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}
$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2\sigma_w^2$. Under this condition,

$$
\mathbb{E}[u_{\mathrm{OVB},i}^2|\mathrm{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{13}
$$

Thus, under our set of assumptions, $\mathbb{E}[u_{\mathrm{OVB},i}^2|\mathrm{x}_i]$ does not depend on x_i and $\mathbb{E}[u_{\mathrm{OVB},i}^2|\mathrm{x}_i] = \mathbb{E}[u_{\mathrm{OVB},i}^2]$. We denote this quantity $\sigma_{u_{\mathrm{OVB}}}^2$.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\mathrm{OVB}}$, noting that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\mathrm{OVB},i}^2] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[u_{\mathrm{OVB},i}^2|\mathrm{x}_i]] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\sigma_{u_{\mathrm{OVB}}}^2] = \sigma_{u_{\mathrm{OVB}}}^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$. And thus $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\mathrm{OVB},i}^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \sigma_{u_{\mathrm{OVB}}}^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Plugin this and equation 12 into equation 11, we get, for $\hat{\beta}_{\mathrm{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\mathrm{OVB}}$:

$$
\hat{\beta}_{\mathrm{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_{u_{\mathrm{OVB}}}^2}{n\ \sigma_x^2}\right)
$$

And substituting for σ_v^2,

$$
\boxed{\hat{\beta}_{\mathrm{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}\right)}
$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{OVB}} &= \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \\ \sigma_{\text{OVB}}^2 &= \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2} = \frac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \sigma_x^2} - \frac{b_{\text{OVB}}^2}{n} \end{aligned} \tag{14}$$

Note, in the absence of omitted variable bias, we retrieve the usual asymptotic distribution of the OLS estimator. To do so, we can:

— Set $\delta = 0$. In this case $b = 0$ and $\sigma_{\text{OVB}}^2 = \dfrac{\sigma_u^2}{n\sigma_x^2}$, the usual variance.

— Set $\sigma_w = 0$ yields the same bias and variance.

— Set $\gamma = 0$. In this case $b = 0$ and $\sigma_{\text{OVB}}^2 = \dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\sigma_x^2}$. The numerator of this variance differs because the variance of y not explained by x includes the effect of w. It is therefore equivalent to the usual variance.

IV. 1.2 Exaggeration ratio and its evolution

Plugging in the values for β_{OVB}, σ_{OVB} and b_{OVB}, from equation 14 into equation 4, we get:

$$\begin{aligned} E_{\text{OVB}} &= 1 + \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} + \frac{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}}{\beta_1}\lambda\left(\frac{\beta_1 + \dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}}{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}}} - z_\alpha\right) \\ &= 1 + \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1 + \dfrac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right) \end{aligned}$$

Keeping in mind that λ is a decreasing function, we notice that E_{ovb} decreases with β_1, σ_x and n and increases with σ_u. Analytically deriving the evolution of this quantity with the omitted variable bias, *ie*, with δ, γ and σ_w is more challenging. The direction of this evolution is likely to depend on the value of the parameters. Since this is not the focus of the present paper, we do not dive into this. We however expect that, if the bias is upward (δ and γ of the same sign), we expect the exaggeration to increase with δ, γ and σ_w^2, for realistic values of the various parameters since the usual omitted variable bias $\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)$ is a major component of the exaggeration.

IV. 2 Instrumental Variable

Now, let's assume that there exists a valid instrumental variable z_i for x_i. We want to estimate an IV regression in which we regress y on x instrumented by z. More precisely, we instrument $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We thus consider a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$.

IV. 2.1 Setting

Assume that $\mu_x + \epsilon_i = \pi_0 + \pi_1 z_i + \epsilon_{\text{IV},i}$. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\text{rank}(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. The data generating process for x_i becomes:

$$x_i = \pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_{\text{IV},i} \tag{15}$$

where $(\pi_0, \pi_1, \gamma) \in \mathbb{R}^3$ are unknown parameters. We assume that $\epsilon_{\text{IV},i}$ is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i \epsilon_{\text{IV},i}] = 0$ and $\mathbb{E}[w_i \epsilon_{\text{IV},i}] = 0$. Note that, since x_i and w_i are correlated, $\gamma \neq 0$. We also assume homoskedasticity for this term, such that $\mathbb{E}[\epsilon^2_{\text{IV},i} | z_i, w_i] = \sigma^2_{\epsilon_{\text{IV}}}$ is constant. It implies that $\mathbb{E}[\epsilon^2_{\text{IV},i} | z_i] = \sigma^2_{\epsilon_{\text{IV}}}$ (since $\mathbb{E}[\epsilon^2_{\text{IV},i} | z_i] = \mathbb{E}[\mathbb{E}[\epsilon^2_{\text{IV},i} | z_i, w_i] | z_i] = \mathbb{E}[\sigma^2_{\epsilon_{\text{IV}}} | z_i] = \sigma^2_{\epsilon_{\text{IV}}}$).

The following Directed Acyclic Graph (DAG) sums up the relationships between the variables. It also helps underline the independence, or not, of some variables.

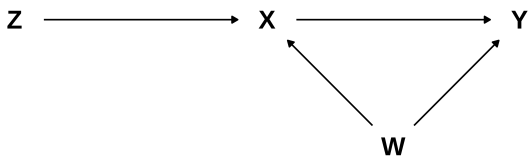

Figure 2 – DAG of the data generating process for the IV

IV. 2.2 Asymptotic distribution

Lemma 3. *Based on the sus-mentioned data generating process, for the IV estimate $\hat{\beta}_{\text{IV}}$ of the regression of y on x instrumented by z, we have:*

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\delta^2 \sigma_w^2 + \sigma_u^2}{\pi_1^2 \sigma_z^2}\right)$$

Proof. We know that we have:

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{IV}} &= (Z'X)^{-1} Z' y \\
&= (Z'X)^{-1} Z' (X\boldsymbol{\beta} + \delta w + u) \\
&= \boldsymbol{\beta} + (Z'X)^{-1} Z' w \delta + (Z'X)^{-1} Z' u
\end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}} | X, Z] = \boldsymbol{\beta} + (Z'X)^{-1} \underbrace{\mathbb{E}[Z'w]}_{0} \delta + (Z'X)^{-1} \underbrace{\mathbb{E}[Z'u]}_{0} = \boldsymbol{\beta} \quad \Rightarrow \quad \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}}] = \boldsymbol{\beta}$$

The IV estimator is unbiased. We now want to derive its asymptotic distribution.

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \delta\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i u_i\right)$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i \mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{z}_i \mathrm{x}_i']$ and since $\mathbb{E}[\mathrm{z}_i w_i] = 0$ and $\mathbb{E}[\mathrm{z}_i u_i] = 0$, the Central Limit Theorem (CLT) gives $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i w_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2])$ and $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{z}_i u_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2])$.

Now, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' u_i^2] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[u_i^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{z}_i)] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[\sigma_u^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \sigma_u^2] = \sigma_u^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. Now, since z_i is a valid instrument, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{z}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i' w_i^2] = \mathbb{E}[\mathrm{z}_i \mathrm{z}_i' \mathbb{E}[w_i^2|\mathrm{z}_i]] = \sigma_w^2 \mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\delta^2\sigma_w^2 + \sigma_u^2)\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)'\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n}\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)'\right)}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i]$, $\mathbb{E}[\mathrm{x}_i \mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i \mathrm{z}_i'] = \mathbb{E}\begin{bmatrix} 1 & z \\ z & z^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i'] = \begin{bmatrix} 1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i] \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[\epsilon_{\text{IV},i}]}_{0} \\ \mu_z & \pi_0\mathbb{E}[z_i] + \pi_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i \epsilon_{\text{IV},i}]}_{0} \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mu_z \\ \mu_z & \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1} = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\pi_0 - \pi_1\mu_z \\ -\mu_z & 1 \end{bmatrix}$$

$$\text{and} \quad \left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\mu_z \\ -\pi_0 - \pi_1\mu_z & 1 \end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i'] = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_1\sigma_z^2 & -\pi_0\sigma_z^2 \\ 0 & \sigma_z^2 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{\pi_0}{\pi_1} \\ 0 & \frac{1}{\pi} \end{bmatrix}$$

And finally,

$$\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} 2\pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) + \frac{\pi_0^2}{\pi_1} & -\mu_z - \frac{\pi_0}{\pi_1} \\ -\mu_z - \frac{\pi_0}{\pi_1} & \frac{1}{\pi_1} \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\boxed{\hat{\beta}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\beta_1,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \pi_1^2\sigma_z^2}\right)} \tag{16}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{IV}} &= 0 \\ \sigma_{\text{IV}}^2 &= \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \pi_1^2\sigma_z^2} \end{aligned} \tag{17}$$

Note that, since z_i is uncorrelated with w_i and $\epsilon_{\text{IV},i}$,

$$\begin{aligned} \sigma_x^2 = \text{var}(x_i) &= \text{var}(\pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_{\text{IV},i}) \\ &= \pi_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_{\epsilon_{\text{IV}}}^2 + 2\pi_1\gamma\underbrace{\text{cov}(z_i, w_i)}_{0} + 2\pi_1\underbrace{\text{cov}(z_i, \epsilon_{\text{IV},i})}_{0} + 2\gamma\underbrace{\text{cov}(w_i, \epsilon_{\text{IV},i})}_{0} \\ &= \underbrace{\pi_1^2\sigma_z^2}_{V_z^x} + \underbrace{\gamma^2\sigma_w^2 + \sigma_{\epsilon_{\text{IV}}}^2}_{V_{\text{non-}z}^x} \end{aligned}$$

For a given variance of x, the larger the proportion of this variance that depends on z, V_z^x, (and the smaller the non-z component $V_{\text{non-}z}^x$), the more precise the estimate.

Also note that,

$$\begin{aligned} & \text{cov}(x_i, z_i) = \text{cov}(\pi_0 + \pi_1 z_i + \gamma w_i + \epsilon_{\text{IV},i}, z_i) = \pi_1\sigma_z^2 \\ \Rightarrow\quad & \rho_{xz} = \text{corr}(x_i, z_i) = \frac{\text{cov}(x_i, z_i)}{\sigma_x\sigma_z} = \pi_1\frac{\sigma_z}{\sigma_x} \\ \Rightarrow\quad & \pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2 \end{aligned}$$

One can therefore rewrite the formula for the variance of our IV estimator as

$$\sigma_{\text{IV}}^2 = \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \sigma_x^2\rho_{xz}^2}$$

This variance formula enables to retrieve the usual result regarding the determinant of the 2SLS variance as discussed in Hansen (2022) for instance. The variance of the estimator increases when σ_u^2, the variance of the error, increases. It decreases with ρ_{xz}, the correlation between x and z, and with σ_x^2, the variance of x, assuming the omitted variable fixed.

IV. 2.3 Exaggeration ratio and its evolution

Plugging in the values for β_{IV}, σ_{IV} and b_{IV}, from equation 17 into equation 4, we get:

$$\begin{aligned} E_{\text{IV}} &= 1 + \frac{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \pi_1^2\sigma_z^2}}}{\beta_1}\lambda\left(\frac{\beta_1}{\sqrt{\dfrac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \pi_1^2\sigma_z^2}}} - z_\alpha\right) \\ &= 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1|\pi_1|\sigma_z\sqrt{n}}\lambda\left(\frac{\beta_1|\pi_1|\sigma_z\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right) \end{aligned}$$

Proposition 1. *E_{IV}, the exaggeration ratio of the instrumental variable estimator, decreases with the strength of the instrument π_1 (or equivalently with the correlation between the endegoneous variable x and the instrument z).*

Proof. Since there is no bias, the only factor varying with the values of the model parameters is the standard error of the estimator. Corollary 1 showed the exaggeration ratio increases with this standard error. Thus, any model parameter positively associated with the later will increase the former. Hence, the exaggeration ratio of the IV estimator, E_{IV}, increases with σ_u, σ_w and δ and decreases with σ_z, π_1 and n. It also decreases with β_1. We thus showed that exaggeration decreases with the strength of the instrument. □

Now, note that:

$$\begin{aligned} \sigma_y^2 &= \text{var}(y_i) = \text{var}(\beta_0 + \beta_1 x_i + \delta w_i + u_i) \\ &= \beta_1^2\sigma_x^2 + \delta^2\sigma_w^2 + \sigma_u^2 + 2\beta_1\delta\underbrace{\text{cov}(x_i, w_i)}_{\gamma\sigma_w^2} + 2\beta_1\underbrace{\text{cov}(x_i, u_i)}_{0} + 2\delta\underbrace{\text{cov}(w_i, u_i)}_{0} \\ &= \beta_1^2\sigma_x^2 + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\ &= \beta_1^2(\pi_1^2\sigma_z^2 + \gamma^2\sigma_w^2 + \sigma_{\epsilon_{\text{IV}}}^2) + (\delta^2 + 2\beta_1\gamma\delta)\sigma_w^2 + \sigma_u^2 \\ &= \underbrace{\beta_1^2\pi_1^2\sigma_z^2}_{V_z^y} + \underbrace{(\beta_1\gamma + \delta)^2\sigma_w^2 + (\beta_1^2\sigma_{\epsilon_{\text{IV}}}^2 + \sigma_u^2)}_{V_{\text{non-}z}^y} \end{aligned}$$

The larger the parameters that are positively associated with E_{IV} (σ_u, σ_w and δ), the larger the proportion of the variance of y that does not depend on z, $V_{\text{non-}z}^y$. Hence, for a given variance of y, the smaller the proportion of this variance that depends on z, V_z^y, (and the larger the non-z component $V_{\text{non-}z}^y$), the less precise the estimate and the larger the exaggeration. The smaller the proportion of the variance in y that depends on z, the larger the exaggeration of the IV estimator.

Since $\pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, we can also express this exaggeration ratio as:

$$E_{\text{IV}} = 1 + \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}} - z_\alpha\right)$$

Note that E_{IV} decreases with ρ_{xz}, the correlation between the endogenous variable x and the instrument z.

Finally, note that if z is a binary variable, we have $\sigma_z = p_1(1-p_1)$ where p_1 the proportion of 1 in the vector z, $\sigma_z^2 = p_1(1-p_1)$ and thus, for all other parameters fixed, including the number of observations n and the true effect size β_1, E_{IV} increases when the proportion of treated observations p_1 gets further from 0.5. $p_1(1-p_1)$ is a parabolic function with maximum reached for $p_1 = 0.5$. When most observations have the same treatment status, *i.e.*, p_1 close to 0 or 1, the σ_z and thus E_{IV} shoot up. There is not enough variation in the treatment status to precisely identify the effect of interest.

IV. 3 Reduced form

Let's now assume that we want to directly estimate the effect of the instrument on the outcome of interest. Such a reduced form approach is often used when the instrument is binary, as described at the end of the IV section. Here, I focus on this binary case but note that most of the analysis carries to a continuous case. I thus consider a reduced form equation in which we regress y on an exogenous shock D:

$$y = D\boldsymbol{\beta}_{\text{RED}} + u_{\text{RED}}$$

where $u_{\text{RED}} = (u_{\text{RED},1}, ..., u_{\text{RED},n})' \in \mathbb{R}^n$ and by the exogeneity assumption $\mathbb{E}[D'u_{\text{RED}}] = 0$.

IV. 3.1 Setting

I also consider a DGP analogous to that of the IV. Since we will directly regress y on z, the resulting estimand will be different. To make them comparable, for the reduced form case, in equation 15, I set π_1 to 1. An increase of 1 in z causes an increase of β_1 in y and the estimands become comparable. For clarity, I denote the discrete shock $\mathrm{d}_i = (1, d_i)'$. All other assumptions and features of the DGP remain unchanged. Under these assumptions, plugging equation 15 into equation 1 yields:

$$y_i = \underbrace{\beta_0 + \beta_1\pi_0}_{\beta_{0,\text{RED}}} + \beta_1 d_i + \underbrace{(\delta + \beta_1\gamma)}_{\delta_{\text{RED}}} w_i + \underbrace{u_i + \beta_1\epsilon_{\text{RED},i}}_{u_{\text{RED},i}}$$

where ϵ_{RED} is equivalent to ϵ_{IV} in the IV. The second component of the parameter vector associated with d_i is thus identical to the second component of $\boldsymbol{\beta}$. For the sake of clarity, we also assume that $\pi_0 = 0$ such that $b_0 = \beta_0$ and thus $(b_0, \beta_1) = \boldsymbol{\beta}'$. This assumption is not necessary but clarifies the analysis. Dropping it does not change the results for the second component of $\boldsymbol{\beta}$, the component of interest. Thus, in the reduced form case, we have:

$$y_i = \mathrm{d}_i'\boldsymbol{\beta} + \delta_{\text{RED}} w_i + u_{\text{RED},i} \tag{18}$$

IV. 3.2 Asymptotic distribution

Lemma 4. *Based on the sus-mentioned data generating process, for the reduced form estimate*

$\hat{\beta}_{\text{RED}}$ of the regression of y on d, we have:

$$\sqrt{n}\left(\hat{\beta}_{\text{RED}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}{\sigma_d^2}\right)$$

with $\sigma_d^2 = p_1(1-p_1)$, p_1 being the proportion of treated observations, i.e., the proportion of 1 in d.

Proof. We first derive the bias of the estimator.

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{\text{RED}} &= (D'D)^{-1}D'y \\ &= (D'D)^{-1}D'(D\boldsymbol{\beta} + \delta_{\text{RED}}w + u_{\text{RED}}) \\ &= \boldsymbol{\beta} + \delta_{\text{RED}}(D'D)^{-1}D'w + (D'D)^{-1}D'u_{\text{RED}}\end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D] = \boldsymbol{\beta} + (D'D)^{-1}\underbrace{\mathbb{E}[D'w]}_{0}\delta_{\text{RED}} + (D'D)^{-1}\underbrace{\mathbb{E}[D'u_{\text{RED}}]}_{\substack{\beta_1\mathbb{E}[D'\epsilon_{\text{RED}}]+\mathbb{E}[D'u] \\ =\ \beta_1 0+0\ =\ 0}} = \boldsymbol{\beta}$$

By the Law of Iterated Expectations, we get that $\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{RED}}|D]] = \beta$ and thus $\hat{\boldsymbol{\beta}}_{\text{RED}}$ is an unbiased estimator of $\boldsymbol{\beta}$. We then derive its asymptotic distribution, noting that:

$$\begin{aligned}\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) &= \left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\mathrm{d}_i'\right)^{-1}\delta_{\text{RED}}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\mathrm{d}_i'\right)^{-1}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i u_{\text{RED},i}\right) \\ &= \left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\mathrm{d}_i'\right)^{-1}\delta_{\text{RED}}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i w_i\right) \\ &\quad + \left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\mathrm{d}_i'\right)^{-1}\left[\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i u_i\right) + \beta_1\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\epsilon_{\text{RED},i}\right)\right]\end{aligned}$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\mathrm{d}_i' \xrightarrow{p} \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$ and since $\mathbb{E}[\mathrm{d}_i w_i] = 0$,$\mathbb{E}[\mathrm{d}_i u_i] = 0$ and $\mathbb{E}[\mathrm{d}_i\epsilon_{\text{RED},i}] = 0$, the Central Limit Theorem (CLT) gives:

$$\begin{aligned}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i w_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2]) \\ \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i u_i\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2]) \\ \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^n \mathrm{d}_i\epsilon_{\text{RED},i}\right) &\xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\epsilon_{\text{RED},i}^2])\end{aligned}$$

As for the IV, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' u_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[u_i^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{d}_i)] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\sigma_u^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_u^2] = \sigma_u^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. In addition, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\epsilon_{\text{RED},i}^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[\epsilon_{\text{RED},i}^2|\mathrm{d}_i]] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\sigma_{\epsilon_{\text{RED}}}^2] = \sigma_{\epsilon_{\text{RED}}}^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$. Now, since d_i is exogenous, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{d}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i' w_i^2] = \mathbb{E}[\mathrm{d}_i\mathrm{d}_i'\mathbb{E}[w_i^2|\mathrm{d}_i]] = \sigma_w^2\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\delta_{\text{RED}}^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2)\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

Now, since $\delta_{\text{RED}} = \delta + \beta_1\gamma$,

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{RED}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, \{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2\}\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1}$.

$$\mathbb{E}[\mathrm{d}_i\mathrm{d}_i'] = \mathbb{E}\begin{bmatrix} 1 & d \\ d & d^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_d \\ \mu_d & \sigma_d^2 + \mu_d^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{d}_i\mathrm{d}_i']^{-1} = \frac{1}{\sigma_d^2}\begin{bmatrix} \sigma_d^2 + \mu_d^2 & -\mu_d \\ -\mu_d & 1 \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{RED}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{RED}}$, we have:

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}{n\sigma_d^2}\right)}$$

□

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{RED}} &= 0 \\ \sigma_{\text{RED}}^2 &= \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}{n\sigma_d^2} \end{aligned} \tag{19}$$

Notice that, as expected, the numerator of the variance corresponds to the part of the variance in y that is orthogonal to d, *i.e.*, the variance of the error term of the regression of y on d:

$$\begin{aligned} \mathrm{Var}\left(y_i^{\perp d}\right) &= \mathrm{Var}\left((\delta + \beta_1\gamma)w_i + u_i + \beta_1\epsilon_{\text{RED},i}\right) \\ &= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2 \\ &\quad + 2(\delta + \beta_1\gamma)\underbrace{\mathrm{cov}(w_i, u_i)}_{0} + 2\beta_1\underbrace{\mathrm{cov}(u_i, \epsilon_{\text{RED},i})}_{0} + 2(\delta + \beta_1\gamma)\beta_1\underbrace{\mathrm{cov}(w_i, \epsilon_{\text{RED},i})}_{0} \\ &= (\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2 \end{aligned}$$

Further, since d is a vector of dummies, denoting p_1 the proportion of 1 in this vector, $\sigma_d^2 = p_1(1-p_1)$ and thus:

$$\hat{\boldsymbol{\beta}}_{\text{RED}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}{n \times p_1(1 - p_1)}\right) \tag{20}$$

As discussed in the IV case, when most observations have the same treatment status, *i.e.*, p_1 close to 0 or 1, the variance shoots up. There is not enough variation in the treatment status to precisely identify the effect of interest.

IV. 3.3 Exaggeration ratio and its evolution

Plugging in the values for β_{RE}, σ_{RE} and b_{RED}, from equation 19 into equation 4, we get:

$$E_{\text{RED}} = 1 + \frac{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}}{\beta_1\sqrt{n \times p_1(1-p_1)}}\lambda\left(\frac{\beta_1\sqrt{n \times p_1(1-p_1)}}{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}} - z_\alpha\right)$$

Proposition 2. *E_{RED}, the exaggeration ratio of the reduced form estimator, increases when the proportion of treated observations moves away from 0.5.*

Proof. Equation 20 showed that σ^2_{RED} increases when the proportion of treated observations moves away from 0.5. Applying corollary 1 directly proves the proposition. □

V Comparison of exaggeration ratios

Now that we have computed the exaggeration ratios of the various estimators, we can study the exaggeration/confounder trade-off in each case. Before doing so, we briefly compare the efficiencies of each estimator.

V. 1 Comparison of efficiencies

The IV and naive regression variances are rather similar. Note that $\sigma_x^2\rho_{xz}^2 < \sigma_x^2$ and,

$$\frac{\gamma^2\sigma_w^2}{\sigma_x^2} > 0 \quad \Leftrightarrow \quad 1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2} < 1 \quad \Leftrightarrow \quad \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) < \sigma_u^2 + \delta^2\sigma_w^2$$

Both these inequalities imply that $\boxed{\sigma^2_{\text{IV}} > \sigma^2_{\text{OVB}}}$. While unbiased, the IV estimator is less precise than the naive estimator. This is a well known and intuitive result that we recovered in a particular setting, tied to the modeling assumptions we made. This result is due to the fact that the IV only uses part of the variation in the endogenous dependent variable, the part that is explained by the instrument.

Focusing on the setting considered for the reduced form estimation, *i.e.*, with $\pi_0 = 0$, $\pi_1 = 1$ and z denoted with d, the asymptotic distribution of the IV estimate in this setting would be:

$$\sqrt{n}\left(\hat{\beta}_{\text{IV(RED)}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{\sigma_d^2}\right)$$

We can notice that $\sigma^2_{\text{RED}} = \sigma^2_{\text{IV(RED)}} + \dfrac{2\beta_1\gamma\sigma_w^2 + \beta_1^2\gamma^2 + \beta_1^2\sigma^2_{\epsilon_{\text{RED}}}}{\sigma_d^2}$. We assumed $\beta_1 > 0$. It makes sense to focus on the case for which $\gamma > 0$ and $\delta > 0$, *i.e.*, the OVB described in equation 14 is an upward bias and causes the same types of issues as the exaggeration. In this case $\sigma^2_{\text{RED}} > \sigma^2_{\text{IV(RED)}}$. In the special setting considered for the reduced form estimation and under these assumptions, we thus have $\boxed{\sigma^2_{\text{RED}} > \sigma^2_{\text{IV(RED)}} > \sigma^2_{\text{OVB(RED)}}}$.

V. 2 The exaggeration/confounders trade-off

The objective of this paper is to compare "biases" resulting from exaggeration and confounders. In this document, we thus aim to compare E_{OVB} to E_{IV} and E_{RED}. We thus define $\Delta_{\text{IV}} = E_{\text{IV}} - E_{\text{OVB}}$ and $\Delta_{\text{RED}} = E_{\text{RED}} - E_{\text{OVB}}$ and analyse whether and for which parameters values these quantities are positive. In particular, we focus on the strength of the IV (or the correlation between the instrument and the endogenous variable) and the proportion of treated observations in the reduced form case.

V. 2.1 Instrumental variable

Proposition 3. *Under the data generating process described in equation 1, for some parameter values, $E_{\text{IV}} > E_{\text{OVB}}$. More specifically, $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$.*

Proof. Combining lemmas 2, 3 and 1, we get:

$$\Delta_{\text{IV}} = E_{\text{IV}} - E_{\text{OVB}} = \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}}\lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}} - z_\alpha\right) - \frac{\delta\gamma\sigma_w^2}{\beta_1\sigma_x^2} - \frac{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1+\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2\left(1-\frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right)$$

We want to show that $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $\Delta_{\text{IV}} > 0$. First, note that only E_{IV} depends on ρ_{xz}. Since E_{IV} decreases with ρ_{xz}, Δ_{IV} also decreases with ρ_{xz}. Now, for fixed and finite values of the other parameters, and since we assumed that $\beta_1 > 0$,

$$\lim_{\rho_{xz}\to 0^+} \frac{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}}{\beta_1\rho_{xz}\sigma_x\sqrt{n}} = +\infty$$
$$\lim_{\rho_{xz}\to 0^+} \lambda\left(\frac{\beta_1\rho_{xz}\sigma_x\sqrt{n}}{\sqrt{\sigma_u^2+\delta^2\sigma_w^2}} - z_\alpha\right) = \lambda(-z_\alpha) \in \mathbb{R}_+$$

Since the other components of Δ_{IV} are finite and do not depend on ρ_{xz}, $\lim_{\rho_{xz}\to 0^+} \Delta_{\text{IV}} = +\infty$. Since Δ_{IV} is a continuous function of ρ_{xz} on $(0, 1]$, for any set of parameter values consistent with our hypotheses, $\exists \rho_{xz} \in (0, 1]$ s.t. $\Delta_{\text{IV}} > 0$, *i.e.* s.t. $E_{\text{IV}} > E_{\text{OVB}}$. Since $\pi_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2$, it is also straigthforward to show, for any set of parameter values consistent with our hypotheses, $\exists \pi_1 \in \mathbb{R}_+^*$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$. □

We thus have shown that, <u>if the correlation between the instrument and the endogenous variable is low, the exaggeration bias of the IV can be larger than that of the "naive" regression, even in the presence of confounders</u>. The trade-off, *ie* the value of Δ, is affected by this correlation, or the strength of the instrument.

Yet, the value and the sign of Δ depend on the value of the other parameters in a non-trivial maner. In a concrete setting, it might therefore be difficult to identify how "low" the correlation has to be for Δ to be positive.

V. 2.2 Reduced form

Proposition 4. *Under the data generating process described in equation 1, for some parameter values,* $E_{\text{RED}} > E_{\text{OVB}}$. *More specifically,* $\exists p_1 \in (0,1)$ *s.t.* $E_{\text{RED}} > E_{\text{OVB}}$.

Proof. For clarity, we focus on a small number of treated observations, *i.e.* p_1 close to 0. The exact same reasoning applies for a small number of non-treated observations, *i.e.* p_1 close to 1 since $x \mapsto x(1-x)$ is symmetric around 0.5 and takes the value 0 in 0 and 1. Combining lemmas 2, 3 and 1, we get:

$$\begin{aligned}\Delta_{\text{RED}} = E_{\text{RED}} - E_{\text{OVB}} = &\frac{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}}{\beta_1\sqrt{n \times p_1(1-p_1)}}\lambda\left(\frac{\beta_1\sqrt{n \times p_1(1-p_1)}}{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}} - z_\alpha\right)\\ &- \frac{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}}{\beta_1\sigma_x\sqrt{n}}\lambda\left(\frac{\sigma_x\sqrt{n}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2}\right)}{\sqrt{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}} - z_\alpha\right)\end{aligned}$$

The proof is analogous to that of proposition 3. We want to show that $\exists p_1 \in (0,1)$ s.t. $\Delta_{\text{RED}} > 0$. First, note that only E_{RED} depends on p_1. Since E_{RED} increases when p_1 gets further away from 0, Δ_{RED} also increases when $p_1 \to 0^+$ (or equivalently $p_1 \to 1^-$). Now, for fixed and finite values of the other parameters, and since we assumed that $\beta_1 > 0$,

$$\begin{aligned}&\lim_{p_1\to 0^+} \frac{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}}{\beta_1\sqrt{n \times p_1(1-p_1)}} = +\infty\\ &\lim_{p_1\to 0^+} \lambda\left(\frac{\beta_1\sqrt{n \times p_1(1-p_1)}}{\sqrt{(\delta+\beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2}} - z_\alpha\right) = \lambda(-z_\alpha) \in \mathbb{R}_+\end{aligned}$$

Since the other components of Δ_{RED} are finite and do not depend on p_1, $\lim_{p_1\to 0^+} \Delta_{\text{RED}} = +\infty$ ($= lim_{p_1\to 1^-}\Delta_{\text{RED}}$). Since Δ_{RED} is a continuous function of p_1 on $(0,1)$, for any set of parameter values consistent with our hypotheses, $\exists p_1 \in (0,1)$ s.t. $\Delta_{\text{RED}} > 0$, *i.e.* s.t. $E_{\text{RED}} > E_{\text{OVB}}$. □

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